

Speakers



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- 
- Introduction
 - What is Secure Computing (seccomp)
 - eBPF - The Cloud Native Observability Platform
 - Seccomp Demo
 - eBPF-based Runtime Security - Kubernetes Identity Aware Syscall Enforcement
 - Tetragon 101
 - SandboxPolicies (Main Components, BPF implementation)
 - Demo
 - Advantages & Disadvantages of both technologies
 - Wrapping up

What is SecComp

Secure Computing (seccomp)

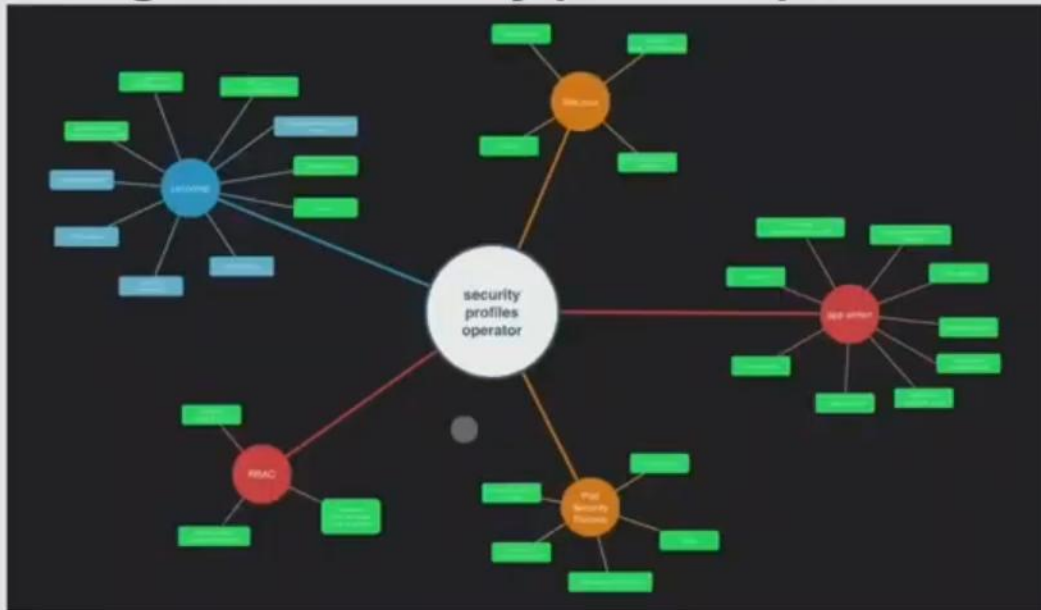
- introduced in 2005 well before many of the primitives for containers.
- Initially introduced to help limit applications that share a kernel to only those system calls needed for the application to run.
- One of the first sandbox mechanisms.
- So many sandboxes out there now from things like firejail all the way to web assembly (wasm)

Secure Computing (seccomp)

- Represents a one way change to secure mode. In which the list of allowed system calls is the constraint for the application.
- Any system call made outside that list will result in an action. You can record an event or kill a process.
- If you want to change the list you need to restart the process.
- The list of allowed system calls has to account for system calls required by container runtimes in our context.

Secure Computing (seccomp)

sigs.k8s.io/security-profiles-operator

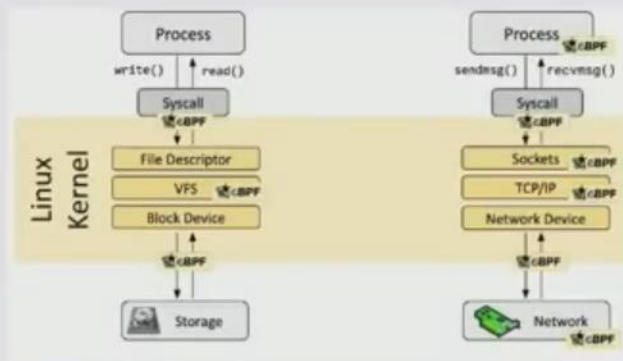


What Is eBPF?

The Cloud Native Observability Platform

eBPF - The Cloud Native Observability Platform

- Extend Linux kernel at most function calls with security and observability context
- Extend Linux kernel to be Kubernetes aware
- Can be dynamically done at scale (10k+ nodes)
- Minimally invasive when done with care
- Extend Linux kernel security model with Kubernetes, metadata and historical context



eBPF - The Cloud Native Observability Platform:

Why is this Interesting?

With an observability platform we can answer...

- **What binaries have run in the past and are running now?**
- **What system calls have been executed from my workload?**
- Are we running latest version of libraries?
- **What network connections are open: both listening, connecting for UDP/TCP/RawSockets/...**
- Is my network healthy: are there drops, is the latency within SLAs, detecting bursts and dips, ...
- **What files are being accessed, written, executed, mmaped, ...**
- **Are my connections encrypted? And with what TLS, IPSec, Wireguard?**
- **Are my TLS connections meeting compliance requirements?**

Competing Ideas

- Real time (time scales of us, ms, ...)
- Minimal CPU and memory constraints
- Offline and Online modes, respect the pipeline limitations

Seccomp Demonstration

Security Profiles Operator

- Huge shoutout to Sascha Grunert and everyone else involved in the project
- The SPO does all of the hard work of generating and pushing security profiles to all nodes.
- The profile needs to be present on disk before the container can start.
- There is a list of system calls that are required for the container runtime do it's part of the work. These are also must be made available to your application.

Security Profiles Operator

- crun requires 65 system calls for just the runtime itself.
- For runc it's 78 system calls.

Security Profiles Operator

- crun requires 65 system calls for just the runtime itself.
- For runc it's 78 system calls.

```
"getpid","getppid","getrlimit","gettid","getuid","keyctl","madvise","mkdirat","mknodat","mmap","mount","mprotect","muommap","nanosleep","newfstatat","open","openat","openat2","pipe2","pivot root","prctl","prlimit64","read","readlinkat","recvfrom","rseq","rt_sigaction","rt_sigprocmask","rt_sigreturn","sched_getaffinity","sched_yield","seccomp","set_robust_list","set_tid_address","setgid","setgroups","sethostname","setuid","setuid","ss_igetitstack","statfs","state","symlinkat","tgkill","umask","umount2","unlinkat","utimensat","write"],"action":"","SCMP_ACT_ALLOW"]}]'
creationTimestamp: "2024-11-15T16:11:03Z"241B
finalizers:
  seccomp-worker-deleted
  seccomp-worker2-deleted
generation: 1
labels:
  spo.x-k8s.io/profile-id: SeccompProfile-runc-v1.2.1
name: runc-v1.2.1
namespace: default
resourceVersion: "788867"
uid: f875649e-331e-4f56-a7a1-c854087034f6
spec:
  architectures:
    SCMP_ARCH_X86_64
  defaultAction: SCMP_ACT_ERRNO
  disabled: false
  syscalls:
    action: SCMP_ACT_ALLOW
    names:
      - arch_prctl
      - brk
```

10,1	Top
11,1	2%
12,1	3%
13,1	4%
14,1	5%
15,1	6%
16,1	7%
17,1	8%
24,1	10%

Security Profiles Operator

- Each runtime uses a different set of syscalls that will be available to your application
- You can further harden this using ebpf!
- You can use seccomp or ebpf or both!

eBPF-based Runtime Security Kubernetes Identity Aware Syscall Enforcement

Kubernetes Identity Aware Syscall Enforcement

Tetragon SandboxPolicies

```
1 apiVersion: cilium.io/v1alpha1
2 kind: SandboxPolicyNamespaced
3 metadata:
4   name: "sensitive-syscall"
5 spec:
6   syscall:
7     - list:
8       - name: "sys_adjtimex"
9       - name: "sys_bpf"
10      - name: "sys_chroot"
11      - name: "sys_clock_gettime"
12      - name: "sys_delete_module"
13      - name: "sys_init_module"
14      - name: "sys_ioprio_load"
15      - name: "sys_madvise"
16      - name: "sys_mount"
17      - name: "sys_pivot_root"
18      - name: "sys_reboot"
19      - name: "sys_seccomp"
20      - name: "sys_getdomainname"
21      - name: "sys_gethostname"
22      - name: "sys_getuid"
23    ...
24    ...
25    op: "In"
26    actions:
27      - type: "Post"
28      - type: "Block"
```

- Tetragon SandboxPolicies
 - Implements Kubernetes Identity Aware Syscall Enforcement (Namespaced policies & Pod label filters)

1. How does it work?
2. What we needed to implement in the BPF side to work correctly?

Tetragon SandboxPolicies - Main Components

1

Declarative Policy
Definition

2

Kubernetes Identity
Aware Selectors

3

Reduced Time Window

Tetragon SandboxPolicies - Declarative Policies (CRD)

```
1 apiVersion: cilium.io/v1alpha1
2 kind: SandboxPolicyNamespaced
3 metadata:
4   name: "sensitive-namespace-syscalls"
5 spec:
6   syscalls:
7     - list:
8       - name: "sys_bpf"
9       - name: "sys_chroot"
10      - name: "sys_clock_settime"
11      - name: "sys_madvise"
12      - name: "sys_mount"
13      - name: "sys_pivot_root"
14      - name: "sys_setdomainname"
15      - name: "sys_sethostname"
16      - name: "sys_setns"
17   op: "In"
18   actions:
19     - type: "Post"
20     - type: "Block"
```

Namespaced Policies
(SandboxPolicyNamespaced)

```
1 apiVersion: cilium.io/v1alpha1
2 kind: SandboxPolicy
3 metadata:
4   name: "sensitive-workload-syscalls"
5 spec:
6   podSelector:
7     matchLabels:
8       "app.kubernetes.io/name": "sensitive-workload"
9   syscalls:
10     - list:
11       - name: "sys_bpf"
12       - name: "sys_chroot"
13       - name: "sys_clock_settime"
14       - name: "sys_madvise"
15       - name: "sys_mount"
16       - name: "sys_pivot_root"
17       - name: "sys_setdomainname"
18       - name: "sys_sethostname"
19       - name: "sys_setns"
20   op: "In"
21   actions:
22     - type: "Post"
23     - type: "Block"
```

**Clusterwide Policies with Pod
label selectors (SandboxPolicy)**

Tetragon SandboxPolicies - JSON event

```
1  { "process_sandbox_syscall": {  
2    "process": {  
3      "cwd": "/root",  
4      "binary": "/usr/bin/mount",  
5      "arguments": ["/dev/sda1 /tmp",  
6      "pod": {  
7        "namespace": "sensitive-namespaces",  
8        "name": "nginx",  
9        "container": {  
10         "id": "containerd://ef365a4bc47a5d618cc5f126aa172970847",  
11         "name": "nginx",  
12         "image": {  
13           "id": "docker.io/library/nginx@sha256:284028649fec70",  
14           "name": "docker.io/library/nginx:latest"  
15         },  
16         "start_time": "2024-11-05T12:34:52Z"  
17       },  
18       "workload": "nginx",  
19       "workload_kind": "Pod"  
20     },  
21     "cap": {  
22       "permitted": {  
23         "CAP_CHOWN",  
24         "DAC_OVERRIDE",  
25         "CAP_SETGID",  
26         "CAP_SETUID",  
27         "CAP_NET_ADMIN",  
28         "CAP_NET_RAW",  
29         "CAP_SYS_CHROOT",  
30         "CAP_SYS_PTRACE",  
31         ...  
32       },  
33     },  
34   },  
35 }
```

```
36   "net": {  
37     "mnt": {  
38       "inode": 4026534398  
39     },  
40     "pid": {  
41       "inode": 4026531836,  
42       "is_host": true  
43     },  
44     "net": {  
45       "inode": 4026531840,  
46       "is_host": true  
47     },  
48     "user": {  
49       "inode": 4026531837,  
50       "is_host": true  
51     },  
52     ...  
53   },  
54   "process_credentials": {  
55     "uid": 0,  
56     "gid": 0,  
57     "euid": 0,  
58     "egid": 0,  
59     "suuid": 0,  
60     "sgid": 0,  
61     "fsuid": 0,  
62     "fsgid": 0  
63   },  
64 }
```

- **Kubernetes Identities** (namespaces, pod labels, workload, workload kind)
- **Process & Parent Information** (binary, arguments, PID etc)
- **Capabilities** (effective, inheritable, permitted)
- **Linux Namespaces** (mount, net, PID etc)
- **Process Credentials** (uid, gid, euid, egid, etc.)
- **Syscall & Policy Information**

```
65   "name": "mount",  
66   "policy": "block-sensitive-syscalls"  
67 },  
68 "node_name": "ip-10-3-5-34.us-west-2.compute.internal",  
69 "time": "2024-11-05T12:44:55.266588256Z",  
70 "cluster_name": "tetragon-dev"  
71 }
```

Tetragon SandboxPolicies - Main Components

1

Declarative Policy
Definition

2

Kubernetes Identity
Aware Selectors

3

Reduced Time Window

Tetragon SandboxPolicies - Kubernetes Identity Aware Selectors

Kubernetes Identity Aware Selectors:

- Namespaced Policies (SandboxPolicyNamespaced)
- Pod label filters (in clusterwide policies)

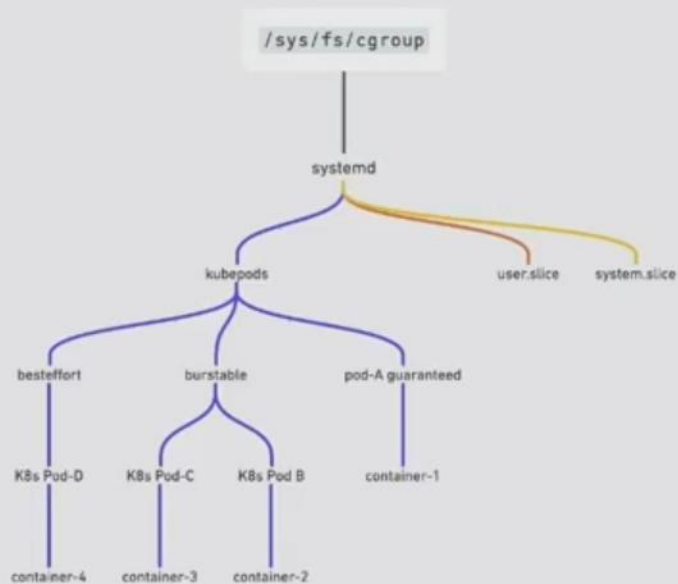
Observability:

- The “In-kernel” filtering is important for K8s Identities for performance reasons (to keep the CPU & memory low) but you *can* live with being the filtering (or at least some of it) in userspace.
- At least corner cases can happen in userspace.

Enforcement:

- Userspace filtering is not an option.
- **The filtering has to happen “In-kernel”, inline with the operation.**
- This means the K8s Identity Aware selectors needs to be mapped to the BPF code.

Tetragon SandboxPolicies - Kubernetes Identity Aware Selectors



Approach: Mapping Kubernetes
Workloads $\Leftrightarrow$ Cgroups

Tetragon SandboxPolicies - Kubernetes Identity Aware Selectors



K8s API
server

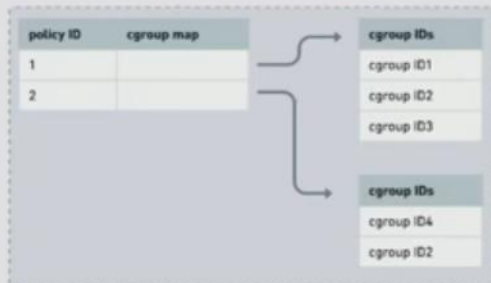


Tetragon
Agent

User space



Linux Kernel



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Tetragon SandboxPolicies - Kubernetes Identity Aware Selectors



K8s API
server



Tetragon
Agent

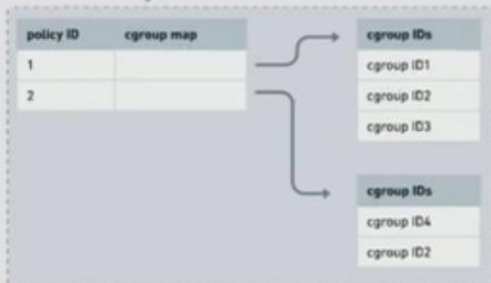
New Policy

Update/Initialise the BPF map

User space



Linux Kernel



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Tetragon SandboxPolicies - Kubernetes Identity Aware Selectors



KBs API
server



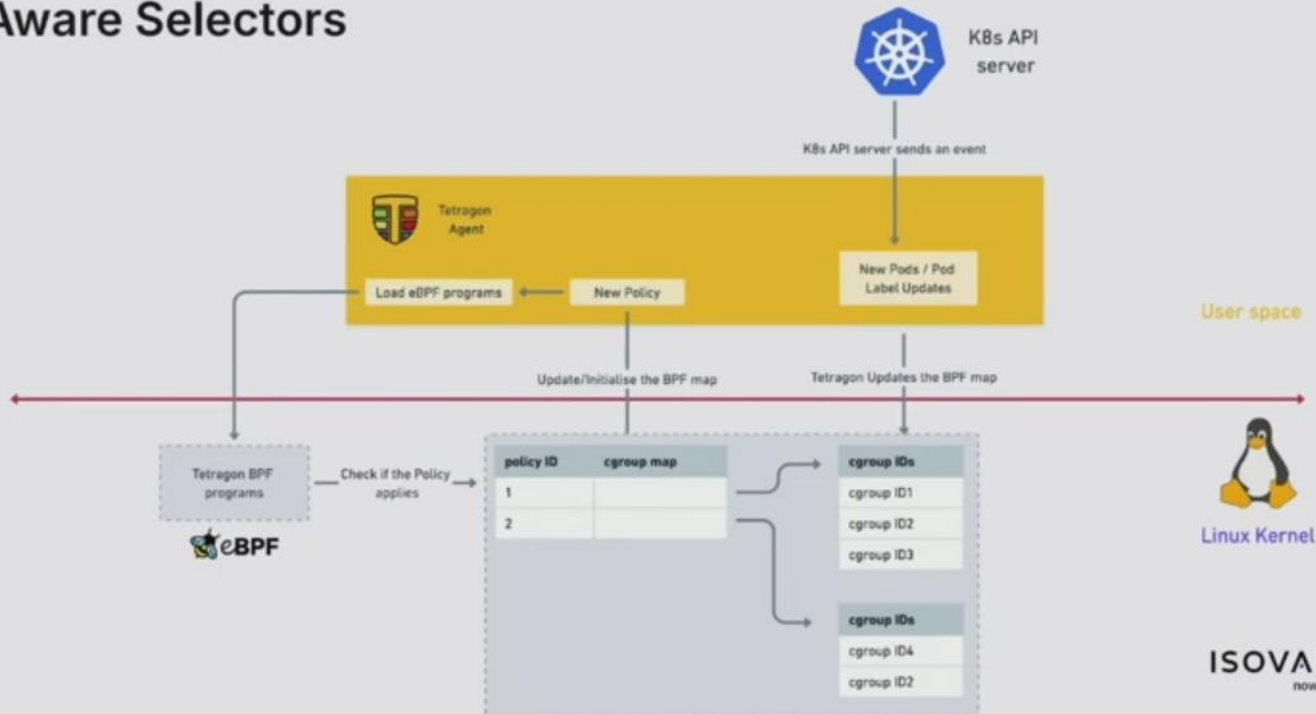
Tetragon SandboxPolicies - Kubernetes Identity Aware Selectors



K8s API
server



Tetragon SandboxPolicies - Kubernetes Identity Aware Selectors



Tetragon SandboxPolicies - Main Components

1

Declarative Policy
Definition

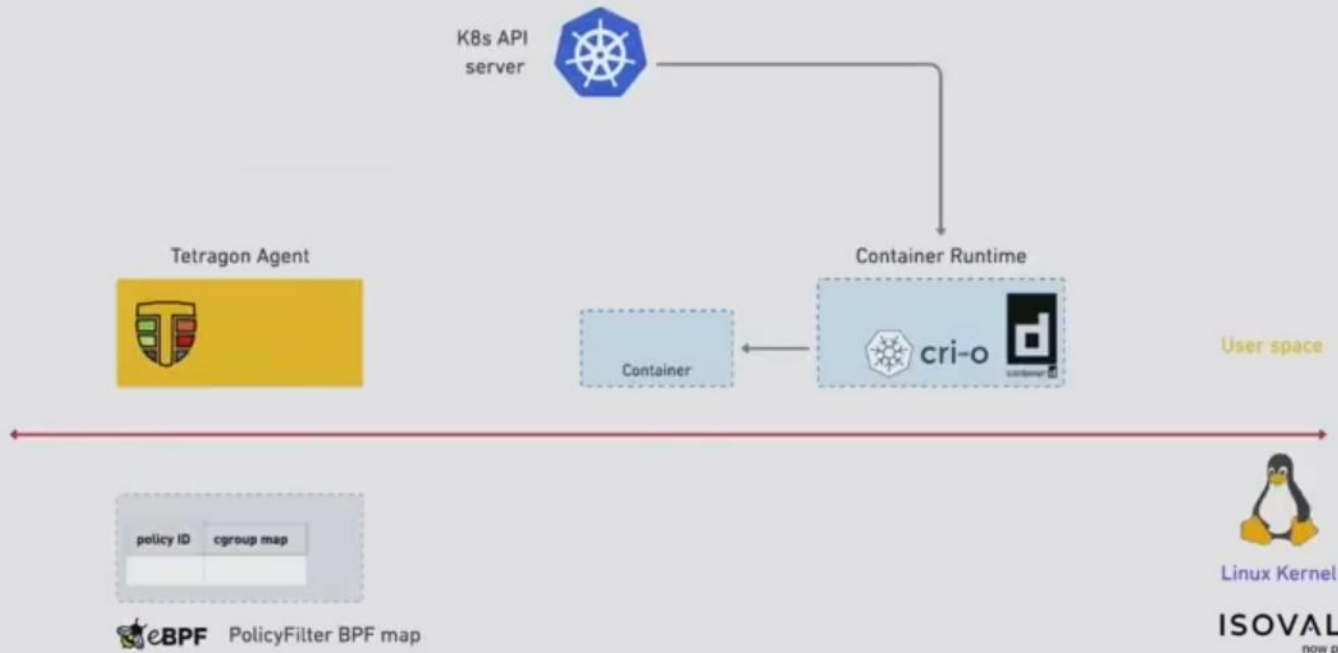
2

Kubernetes Identity
Aware Selectors

3

Reduced Time Window

Tetragon SandboxPolicies - Reduced Time Window



Tetragon SandboxPolicies - Reduced Time Window

Solution: Configuring Runtime Hooks

Tetragon Agent



Container Runtime



User space



PolicyFilter BPF map

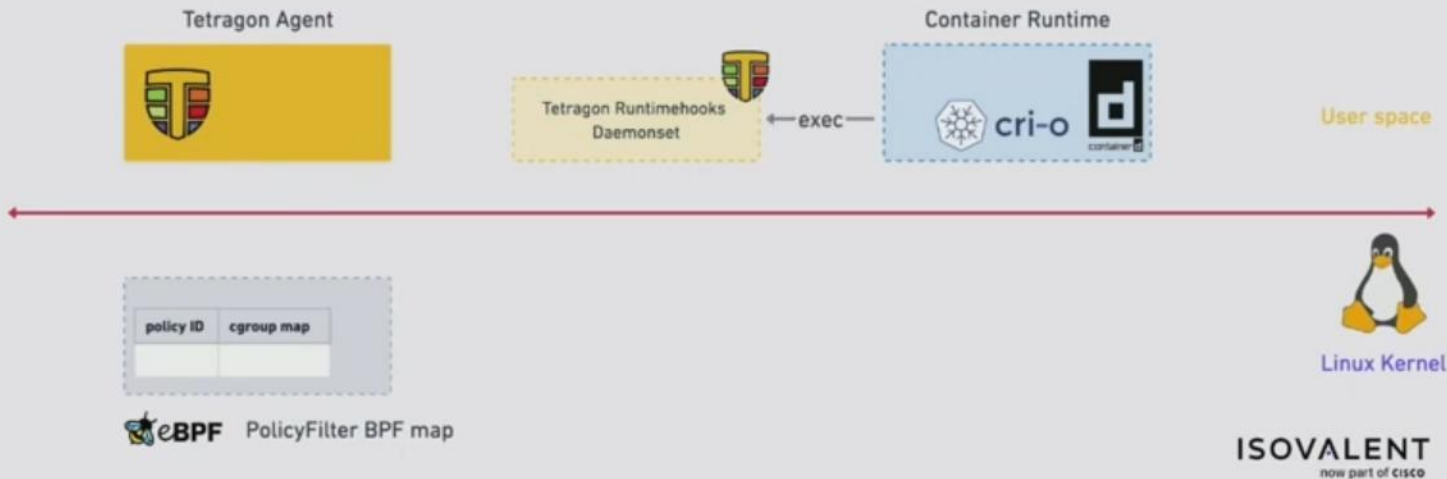


Linux Kernel

ISOVALENT
now part of CISCO

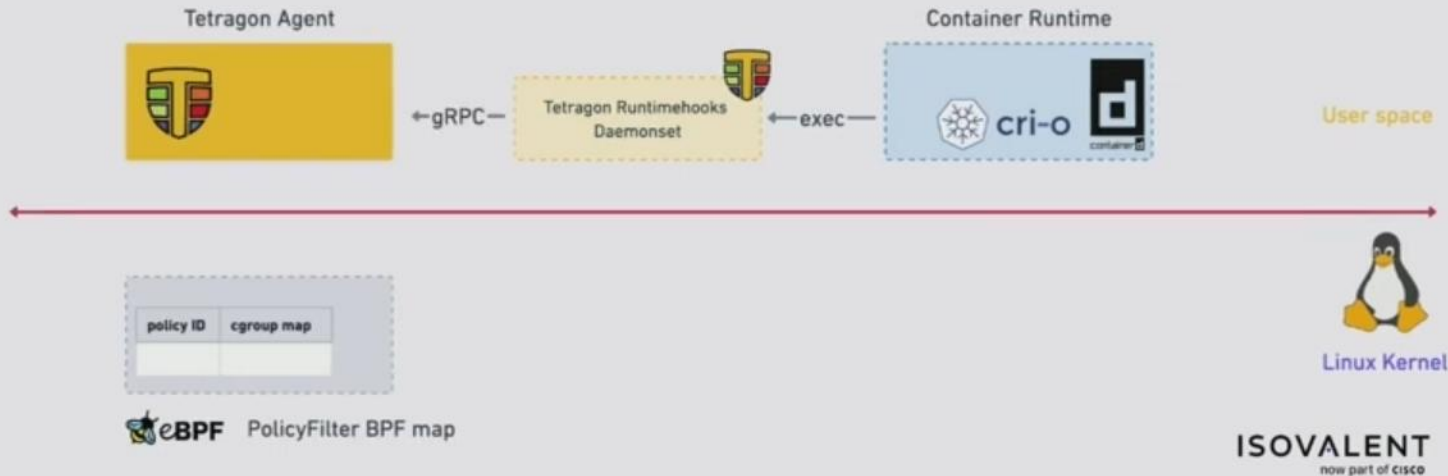
Tetragon SandboxPolicies - Reduced Time Window

Solution: Configuring Runtime Hooks



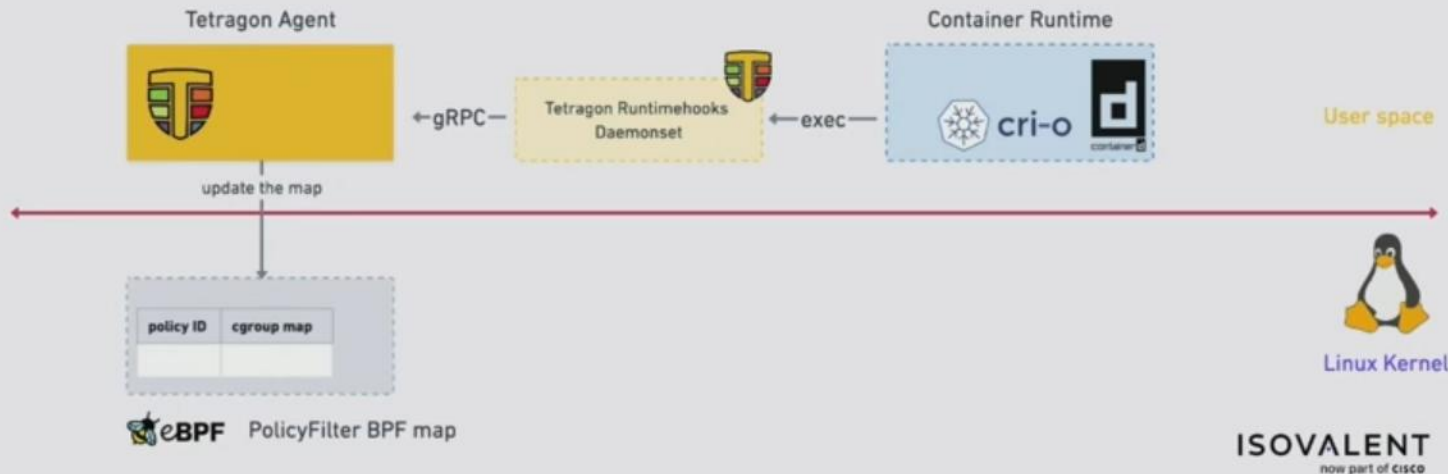
Tetragon SandboxPolicies - Reduced Time Window

Solution: Configuring Runtime Hooks



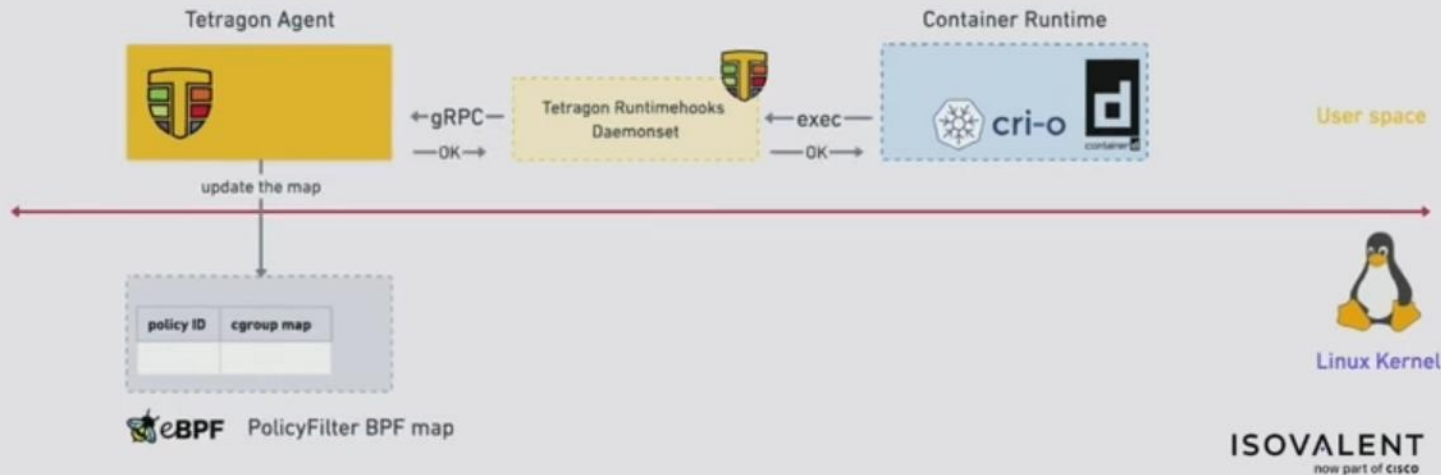
Tetragon SandboxPolicies - Reduced Time Window

Solution: Configuring Runtime Hooks



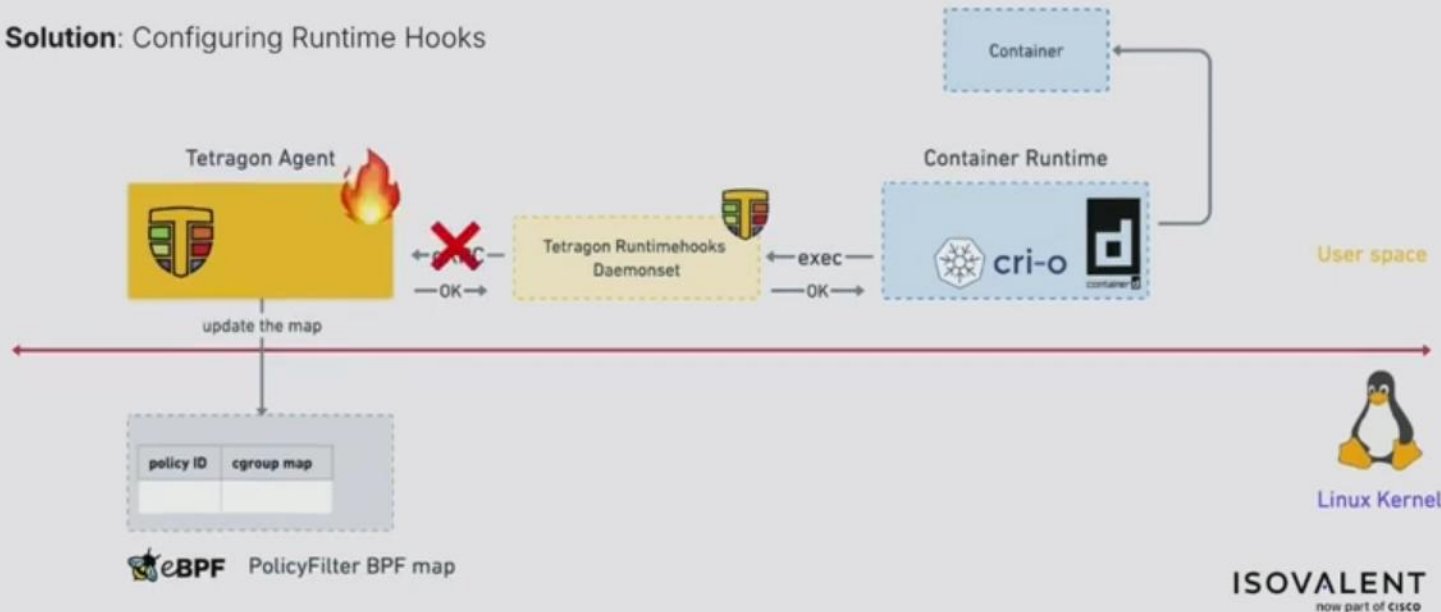
Tetragon SandboxPolicies - Reduced Time Window

Solution: Configuring Runtime Hooks



Tetragon SandboxPolicies - Reduced Time Window

Solution: Configuring Runtime Hooks



Demo - Kubernetes Identity Aware "seccomp"

Demo Environment

Test Environment:

- AWS cluster, 10 nodes, Amazon Linux 2023.6.20241010, 6.1.112-122.189
 - Tetragon (daemonset)
 - Tetragon-rthooks (daemonset)
 - "sensitive-namespace" Kubernetes namespace
 - nginx Kubernetes Pod
 - "sensitive-syscalls" Tetragon SandboxPolicy

Demo steps:

1. Apply "sensitive-syscalls" Tetragon SandboxPolicy on "sensitive-namespace" Kubernetes namespace
2. "kubectl exec" into nginx Kubernetes Pod, invoke sensitive syscalls.

~/sandbox_policies_seccomp_talk

natalia@NATALIA-M-GWPY: ~/sandbox_policies_seccomp_talk

\$ kubectl get pods --n sensitive-namespace --o wide

[18:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23m	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	compa	storage

natalia@NATALIA-M-GWPY: ~/sandbox_policies_seccomp_talk

[18:49:45]

\$

~/sandbox_policies_seccomp_talk

natalia@NATALIA-M-GWPY: ~/sandbox_policies_seccomp_talk

\$

[18:48:29]

~/sandbox_policies_seccomp_talk

ip-10-3-7-230.us-west-2

Q: ip-10-3-7-230.us-west-2

Q: ip-10-3-7-230.us-west-2

```
nat@ps@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk
$ kubectl get pods -n sensitive-namespace -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

```
nat@ps@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk
$
```

```
nat@ps@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk
$
```

```
ip-10-3-7-230.us-west-2
```

_seccomp_talk (-zsh)

natalia@NATALIA-M-6WFF: ~/sandbox_policies_seccomp_talk

\$ kubectl get pods -n sensitive-namespace -o wide

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

natalia@NATALIA-M-6WFF: ~/sandbox_policies_seccomp_talk

\$

[16:49:00]

_seccomp_talk (-zsh)

apiVersion: cilium.io/v1alpha1

kind: SandboxPolicyNamespaced

metadata:

name: "sensitive-syscalls"

spec:

syscalls:

- list:

- name: "sys_adjtimex"
- name: "sys_bpf"
- name: "sys_chroot"
- name: "sys_clock_settime"
- name: "sys_delete_module"
- name: "sys_init_module"
- name: "sys_kexec_load"
- name: "sys_madvise"
- name: "sys_mount"
- name: "sys_pivot_root"

Q~ ip-10-3-7-230.us-we

< > x

kubectl (kubectl)

ip-10-3-7-230: /#

Q~ ip-10-3-7-230.us 3/4

< > x

~ seccomp_talk (-zsh)

```
nataliva@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk
$ kubectl get pods -n sensitive-namespace -o wide
```

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED	NODE	READINESS	GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230-us-west-2.compute.internal	<none>		<none>	

```
nataliva@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk
$
```

[16:49:06]

~ seccomp_talk (-zsh)

```
- name: "sys_reboot"
- name: "sys_seccomp"
- name: "sys_setdomainname"
- name: "sys_sethostname"
- name: "sys_setns"
- name: "sys_settimeofday"
- name: "sys_setxattr"
- name: "sys_stime"
- name: "sys_swapoff"
- name: "sys_swapon"
- name: "sys_unshare"
op: "In"
actions:
  - type: "Post"
  - type: "Block"
```

Q~ ip-10-3-7-230.us-we

< > x

```
nataliva@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk
$
```

[16:49:24]

~ kubectl (kubectl)

ip-10-3-7-230.us

Q~ ip-10-3-7-230.us 3/4

< > x

seccomp_talk (~25h)

natalia@NATALIA-M-6WYF: ~/sandbox_policies_seccomp_talk

\$ kubectl get pods -n sensitive-namespace -o wide

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

natalia@NATALIA-M-6WYF: ~/sandbox_policies_seccomp_talk

\$

[16:49:06]

seccomp_talk (~25h)

```
- name: "sys_settimeofday"
- name: "sys_setxattr"
- name: "sys_time"
- name: "sys_swapoff"
- name: "sys_swapon"
- name: "sys_unshare"
op: "In"
actions:
  - type: "Post"
  - type: "Block"
```

Q~ ip-10-3-7-230.us-we < > x

natalia@NATALIA-M-6WYF: ~/sandbox_policies_seccomp_talk

\$ kubectl apply -f sensitive-syscalls.yaml -n sensitive-namespace

[16:49:24]

sandboxpolicynamespace.cilium.io/sensitive-syscalls created

natalia@NATALIA-M-6WYF: ~/sandbox_policies_seccomp_talk

\$

[16:50:02]

kubectl (kubert@)

ip-10-3-7-230: /#

Q~ ip-10-3-7-230.us 3/4 < > x

nataliva@NATALIVA-M-6WFF:~/sandbox_policies_seccomp_talk

seccomp_talk (-bash)

```
nataliva@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk
$ kubectl get pods -n sensitive-namespace -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

```
nataliva@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk
$
```

[16:48:15]

seccomp_talk (-bash)

```
op: "In"
actions:
  - type: "Post"
  - type: "Block"
```

nataliva@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk

\$ kubectl apply -f sensitive-syscalls.yaml -n sensitive-namespace

sandboxpolicynamespace.cilium.io/sensitive-syscalls created

nataliva@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk

\$ kubectl get sandboxpolicynamespace -n sensitive-namespace

NAME	AGE
sensitive-syscalls	11s

nataliva@NATALIVA-M-6WFF: ~/sandbox_policies_seccomp_talk

\$

[16:49:24]

[16:50:02]

[16:50:13]

kubectl (-bash)

ip-10-3-7-230: /#

Q~ ip-10-3-7-230.us-we

Q~ ip-10-3-7-230.us 3/4

seccomp_talk (-bash)

```

nataiva@NATALIVA-M-GRFY: ~/sandbox_policies/seccomp_talk
$ kubectl get pods -n sensitive-namespace -o wide

```

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

```

nataiva@NATALIVA-M-GRFY: ~/sandbox_policies/seccomp_talk
$

```

[16:49:06]

seccomp_talk (-bash)

```

op: "In"
actions:
  - type: "Post"
  - type: "Block"

```

Q~ ip-10-3-7-230.us-we < > x

```

nataiva@NATALIVA-M-GRFY: ~/sandbox_policies/seccomp_talk
$ kubectl apply -f sensitive-syscalls.yaml -n sensitive-namespace

```

[16:49:24]

```

sandboxpolicynamespace.cilium.io/sensitive-syscalls created

```

```

nataiva@NATALIVA-M-GRFY: ~/sandbox_policies/seccomp_talk
$ kubectl get sandboxpolicynamespace -n sensitive-namespace

```

[16:50:02]

NAME	AGE
sensitive-syscalls	11s

```

nataiva@NATALIVA-M-GRFY: ~/sandbox_policies/seccomp_talk
$

```

[16:50:13]

kubect (kubect)

```

ip-10-3-7-230:~# tetra.getevents -o compact --pods nginx

```

Q~ ip-10-3-7-230.us 3/4 < > x

~/sandbox_policies_seccomp_talk [~]\$

nataliva@NATALIVA-M-6WFY: ~/sandbox_policies_seccomp_talk

\$ kubectl get pods -n sensitive-namespace -o wide

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10-3-7-230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

nataliva@NATALIVA-M-6WFY: ~/sandbox_policies_seccomp_talk

\$ kubectl exec -it -n sensitive-namespace nginx -- /bin/bash

[16:49:06]

~/sandbox_policies_seccomp_talk [~]\$

```
op: "In"
actions:
  - type: "Post"
  - type: "Block"
```

Q~ ip-10-3-7-230.us-we < > x

nataliva@NATALIVA-M-6WFY: ~/sandbox_policies_seccomp_talk

\$ kubectl apply -f sensitive-syscalls.yaml -n sensitive-namespace

[16:49:24]

sandboxpolicynamespace.cilium.io/sensitive-syscalls created

nataliva@NATALIVA-M-6WFY: ~/sandbox_policies_seccomp_talk

\$ kubectl get sandboxpolicynamespace -n sensitive-namespace

[16:50:02]

NAME	AGE
sensitive-syscalls	11s

nataliva@NATALIVA-M-6WFY: ~/sandbox_policies_seccomp_talk

\$

[16:50:13]

~/sandbox_policies_seccomp_talk [~]\$

ip-10-3-7-230:~# tetra getevents -o compact --pods nginx

Q~ ip-10-3-7-230.us 3/4 < > x

```

nataiva@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk
$ kubectl get pods -n sensitive-namespaces -o wide

```

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	amazon	amazon

```

nataiva@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk
$ kubectl exec -it -n sensitive-namespaces nginx -- /bin/bash

```

[16:48:00]

```

root@ip-10-3-7-230:/#

```

```

- sensitive-syscalls.yaml

```

```

op: "In"
actions:
  - type: "Post"
  - type: "Block"

```

Q~ ip-10-3-7-230.us-we < > x

```

nataiva@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk
$ kubectl apply -f sensitive-syscalls.yaml -n sensitive-namespaces

```

[16:49:24]

```

sandboxpolicynamespace.cilium.io/sensitive-syscalls created

```

```

nataiva@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk
$ kubectl get sandboxpolicynamespace -n sensitive-namespaces

```

[16:50:02]

NAME	AGE
sensitive-syscalls	11s

```

nataiva@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk
$

```

[16:50:13]

```

$ kubectl (subshell)

```

```

ip-10-3-7-230:/# tetra getevents -o compact --pods nginx

```

Q~ ip-10-3-7-230.us 3/4 < > x

```

sensitive-namespaces/nginx /bin/bash

```

CAP_SYS_ADMIN

kubectll (kubectll)

natalia@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk

\$ kubectl get pods -n sensitive-namespace -o wide

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

natalia@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk

\$ kubectl exec -it -n sensitive-namespace nginx -- /bin/bash

[16:49:06]

root@ip-10-3-7-230:/# chroot /

chroot: cannot change root directory to '/': Operation not permitted

root@ip-10-3-7-230:/#

~/seccomp_talk (-csh)

op: "In"

actions:

- type: "Post"

- type: "Block"

Q~ ip-10-3-7-230.us-we < > x

natalia@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk

\$ kubectl apply -f sensitive-syscalls.yaml -n sensitive-namespace

[16:49:24]

sandboxpolicynamespace.cilium.io/sensitive-syscalls created

natalia@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk

\$ kubectl get sandboxpolicynamespace -n sensitive-namespace

[16:50:02]

NAME	AGE
sensitive-syscalls	11s

natalia@NATALIA-M-GRFY: ~/sandbox_policies_seccomp_talk

\$

[16:50:13]

kubectll (kubectll)

ip-10-3-7-230:/# kubectl get events -o compact --pods nginx

Q~ ip-10-3-7-230.us 3/4 < > x

✓	Warning	sensitive-namespace/nginx	/bin/bash	✓	CAP_SYS_ADMIN
✓	Warning	sensitive-namespace/nginx	/usr/sbin/crond	✓	CAP_SYS_ADMIN
✓	Warning	sensitive-namespace/nginx	/usr/sbin/crond	sys_chroot (sensitive-syscalls)	CAP_SYS_ADMIN
✓	Warning	sensitive-namespace/nginx	/usr/sbin/crond	✓	CAP_SYS_ADMIN

kubect (kubect)

natali@NATALIA-M-6RFY: ~/sandbox_policies_seccomp_talk

\$ kubectl get pods -n sensitive-namespace -o wide

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

natali@NATALIA-M-6RFY: ~/sandbox_policies_seccomp_talk

\$ kubectl exec -it -n sensitive-namespace nginx -- /bin/bash

[16:49:00]

root@ip-10-3-7-230:/# chroot /

chroot: cannot change root directory to '/': Operation not permitted

root@ip-10-3-7-230:/# mount /dev/sda1 /tmp

mount: /tmp: permission denied.

dmesg(1) may have more information after failed mount system call.

root@ip-10-3-7-230:/#

~# sensitive_syscalls [csh]

op: "In"

actions:

- type: "Post"

- type: "Block"

natali@NATALIA-M-6RFY: ~/sandbox_policies_seccomp_talk

\$ kubectl apply -f sensitive-syscalls.yaml -n sensitive-namespace

[16:49:24]

sandboxpolicynamespace.cilium.io/sensitive-syscalls created

natali@NATALIA-M-6RFY: ~/sandbox_policies_seccomp_talk

\$ kubectl get sandboxpolicynamespace -n sensitive-namespace

[16:50:02]

NAME	AGE
sensitive-syscalls	11s

natali@NATALIA-M-6RFY: ~/sandbox_policies_seccomp_talk

\$

[16:50:13]

kubect (kubect)

ip-10-3-7-230:/# tetra getevents -o compact --pods nginx

Pod	Container	Path	Operation	Result
sensitive-namespace/nginx	/bin/bash		CAP_SYS_ADMIN	Allowed
sensitive-namespace/nginx	usr/sbin/chroot /		CAP_SYS_ADMIN	Allowed
sensitive-namespace/nginx	usr/sbin/chroot sys_chroot (sensitive-syscalls)		CAP_SYS_ADMIN	Allowed
sensitive-namespace/nginx	usr/sbin/chroot /		CAP_SYS_ADMIN	Allowed
sensitive-namespace/nginx	usr/sbin/chroot /dev/sda1 /tmp		CAP_SYS_ADMIN	Allowed
sensitive-namespace/nginx	usr/sbin/chroot /dev/sda1 /tmp		CAP_SYS_ADMIN	Allowed

Q~ ip-10-3-7-230.us-we < > x

Q~ ip-10-3-7-230.us-we < > x

```

natali@NATALIVA-M-GRFY: ~/sandbox_policies_seccomp_talk
$ kubectl get pods -n sensitive-namespace -o wide

```

[16:48:15]

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
nginx	1/1	Running	0	23h	10.3.7.230	ip-10-3-7-230.us-west-2.compute.internal	<none>	<none>

```

natali@NATALIVA-M-GRFY: ~/sandbox_policies_seccomp_talk
$ kubectl exec -it -n sensitive-namespace nginx -- /bin/bash

```

[16:49:06]

```

root@ip-10-3-7-230:/# chroot /
chroot: cannot change root directory to '/': Operation not permitted
root@ip-10-3-7-230:/# mount /dev/sdal /tmp
mount: /tmp: permission denied.
dmesg(1) may have more information after failed mount system call.
root@ip-10-3-7-230:/#

```

```

...seccomp_talk - 137y

```

```

op: "In"
actions:
  - type: "Post"
  - type: "Block"

```

Q~ ip-10-3-7-230.us-we < > x

```

natali@NATALIVA-M-GRFY: ~/sandbox_policies_seccomp_talk
$ kubectl apply -f sensitive-syscalls.yaml -n sensitive-namespace

```

[16:49:24]

```

sandboxpolicynamespace.cilium.io/sensitive-syscalls created

```

```

natali@NATALIVA-M-GRFY: ~/sandbox_policies_seccomp_talk
$ kubectl get sandboxpolicynamespace -n sensitive-namespace

```

[16:50:02]

NAME	AGE
sensitive-syscalls	11s

```

natali@NATALIVA-M-GRFY: ~/sandbox_policies_seccomp_talk
$

```

[16:50:13]

```

kubecti (kubecti)

```

```

ip-10-3-7-230:/# tetra getevents -o compact --pods nginx

```

Q~ ip-10-3-7-230.us-we < > x

Pod	Container	Path	Operation	Result	Cap
sensitive-namespace/nginx	/bin/bash			✓	CAP_SYS_ADMIN
sensitive-namespace/nginx	/usr/sbin/chroot /			✓	CAP_SYS_ADMIN
sensitive-namespace/nginx	/usr/sbin/chroot: sys_chroot (sensitive-syscalls)			✓	CAP_SYS_ADMIN
sensitive-namespace/nginx	/usr/bin/chroot /			✓	CAP_SYS_ADMIN
sensitive-namespace/nginx	/usr/bin/mount /dev/sdal /tmp			✓	CAP_SYS_ADMIN
sandbox sensitive-namespace/nginx	/usr/bin/mount sys.mount (sensitive-syscalls)			✗	CAP_SYS_ADMIN
sensitive-namespace/nginx	/usr/bin/mount: /dev/sdal /tmp			✓	CAP_SYS_ADMIN



Advantages And Disadvantages Of Both Technology

eBPF-based Kubernetes Identity Aware Syscall Enforcement

Advantages:

1. Dynamic

- a. The K8s application doesn't need to be restarted for the policy to take effect
- b. The underlying K8s node doesn't need to be rebooted

2. Flexible - Powerful "in-kernel" selectors

- a. Kubernetes Identity Aware metadata (Pod labels, namespaces)
- b. Binaries & hashes (SHA256)
- c. Process capabilities, Linux namespaces, UNIX credentials

Disadvantages:

- 1. **BTF metadata** file needs to be present
- 2. **Newer kernel (6.1+)** for advanced features (multi-kprobe support)

Seccomp

Advantages:

1. Starts with the application.

- a. There is no window of time that the process is not protected.

2. Good Tooling

- a. Kubernetes supports seccomp by default and the security profile operator is very good.

Disadvantages:

1. Hard to iterate on a seccomp policy without restarting the application.
2. You need to allow for the container runtime syscalls as well as those of your app.

Thank you! Q&A

 [cilium/tetragon](https://github.com/cilium/tetragon)

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